Heurion Clinical AI Workstation

Turn every encounter, file, and confirmation into auditable, evolving, exportable clinical memory.

A self-evolving AI workstation for oncology research and clinical teams. It distills fragmented conversations, records, and documents into a structured, unified Memory Graph.

System architecture: from input to memory loop

Clinical input →	EventLog →	Evolution Engine →	Memory Graph →	Smart output
Chat / Files / Records	Immutable & auditable	Extract · Link · Synthesize	Facts / Articles / Gaps	Search / Reports / Decisions

Async evolution pipeline

INGEST →	EXTRACT →	LINK →	SYNTHESIZE →	CURATE →	RETRIEVE
Append to immutable EventLog	LLM extracts Facts	Deduplicate & link entities	Related Facts → Article	Propagate user edits	Semantic + graph recall

Who is it for

- Oncology researchers: managing enrolled patients, protocols, follow-up data, and adverse events.
- Clinicians / MDT teams: longitudinal patient context, fewer repetitive questions, lower hallucination risk.
- Medical research assistants: producing case summaries, academic slides, Table 1, and statistical charts.
- Hospital / department IT: AI that is explainable, auditable, and deployable on premises.

Three big pain points

- Stateless: bots “forget” after every conversation; doctors repeat the background → wasted time, poor UX.
- Un-auditable: no traceability for conclusions or stale evidence → medical & compliance risk.
- Siloed: chat, records, and reports are disconnected → rework, information islands.

Heurion solution

- Unified Memory Graph: Facts / Articles / Gaps / Skills / Entities / Documents live as nodes and relations.
- Versioned: every edit creates a new version; old versions remain traceable.
- Impact analysis: changing a Fact warns which Articles are affected; underlying changes trigger stale markers and one-click regeneration.

Three core capabilities

1. Unified Memory Graph

Facts, Articles, Gaps, Skills, Entities, and Documents live as nodes on the same graph, connected by relations.

- Versioned & impact-aware: every edit leaves a trail and shows dependent Articles.
- Patient subgraph: focus on nodes related to a specific patient to support MDT decisions.
- Exportable: full archives in .hma (Heurion Memory Archive) format.

2. Evolving Knowledge Base

Knowledge grows and self-corrects through use, not configured once.

- Auto-extract: structured Facts from chat and files.
- Auto-synthesize: Articles generated when enough related Facts accumulate.
- Knowledge Gap: explicitly records “I don't know,” waiting for validation instead of hallucinating.

3. Smart Report Assistant

Turn a clinical discussion into a deliverable document with one sentence — without blocking chat.

- Generate DOCX, PPTX, tables (Table 1, AE summaries), and plots (KM curves, forest plots).
- File links are persisted; refresh the page and the download link still works.
- Generated content can be added to the knowledge base with one click.

Typical use cases

Longitudinal patient management

A lung-cancer patient's CTs, pathology, and treatment history are stored as Facts. The system recalls the patient subgraph instead of giving generic advice.

Protocol writing

A team uploads protocols and literature. The system extracts Facts and synthesizes Articles; one sentence generates the academic PPT and Table 1.

Knowledge maintenance & audit

A manager edits a Fact; the system warns that 3 Articles will be affected, marks them stale, and regenerates them in one click.

Security & compliance

- Tenant isolation: each user's EventLog, facts, knowledge, and files are stored in a separate workspace.
- Two-plane isolation: Control Plane and Execution Plane are separated.
- Auditable: immutable EventLog records every chat, file generation, and fact change.
- Self-host friendly: run everything via Docker Compose on premises or in a private cloud.

Business value

Efficiency: Less time re-supplying patient background; faster case summaries and reports.

Quality: Reduced AI hallucinations and outdated recommendations.

Compliance: Every conclusion traces back to source Facts and versions.

Assetization: Conversations and files become exportable, migratable knowledge assets.